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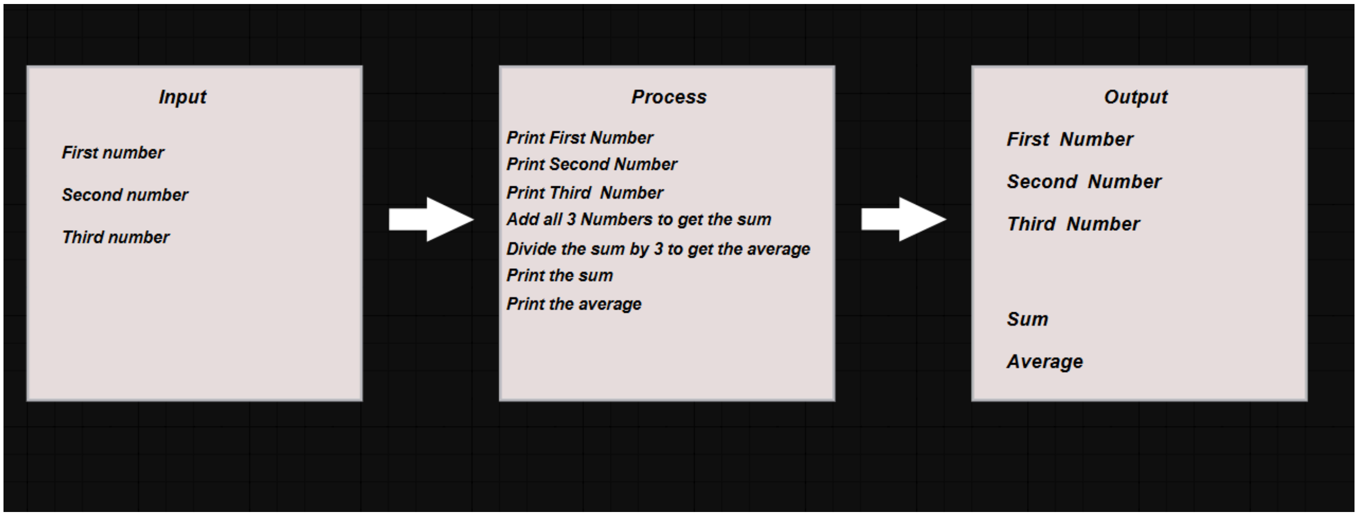
HIPO CHART



NARRATIVE DESCRIPTION

Problem :

Create a program that accepts three input numbers and computes the **sum** and **average** of those numbers. Then, print the original or the three numbers, the calculated **sum**, and the **average**.

Solution

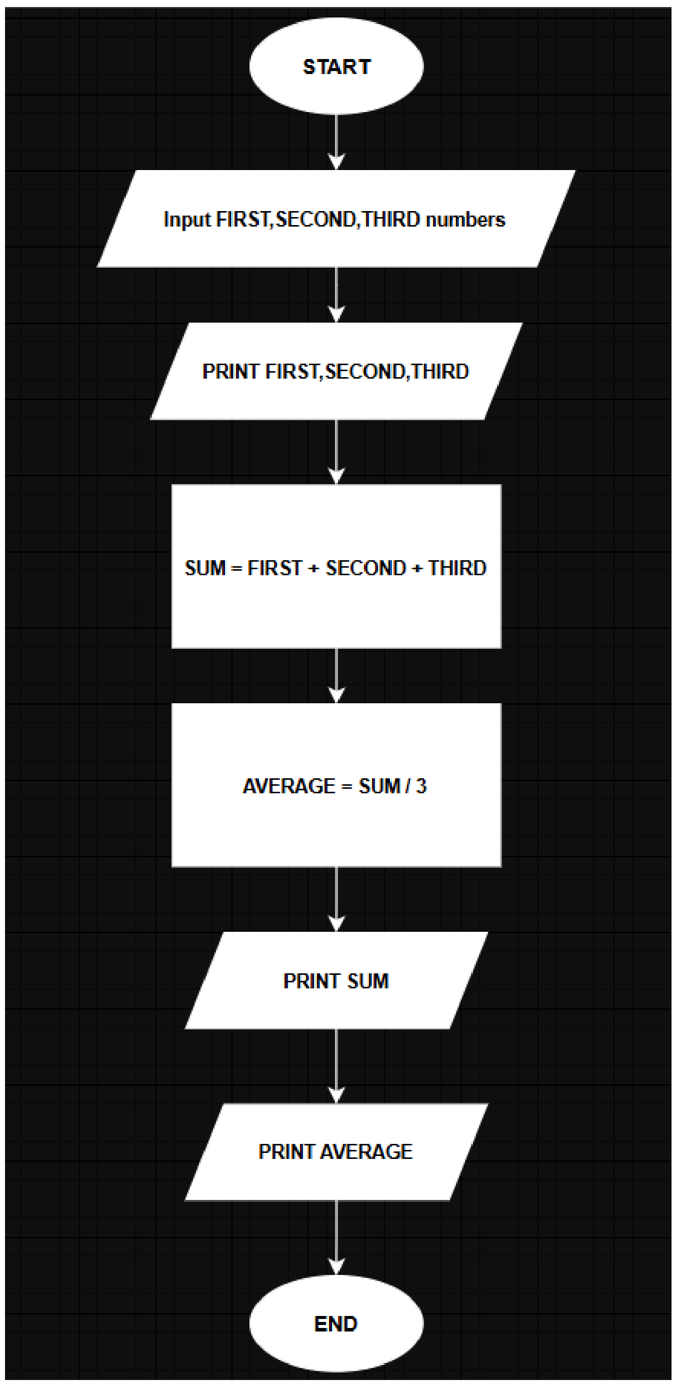
Design in narrative form:

Step 1 : Read the first number and store it in variable FIRST.
Step 2 : Read the second number and store it in variable SECOND.
Step 3 : Read the third number and store it in variable THIRD.
Step 4 : Print the FIRST number.
Step 5 : Print the SECOND number.
Step 6 : Print the THIRD number.
Step 7 : Compute the sum by adding FIRST + SECOND + THIRD and store it in SUM.
Step 8 : Compute the average by dividing SUM by 3 and store it in AVERAGE.
Step 9 : Print the SUM.
Step 10 : Print the AVERAGE.
Step 11 : End the Program.

Explanation :

The program begins by reading three numbers from the user input and storing them in variables FIRST, SECOND, and THIRD. Then these values are then printed to display the original inputs. Then the program computes the SUM by adding the three values FIRST, SECOND, and THIRD. Then calculate the AVERAGE by dividing the SUM by 3. Finally, the SUM and AVERAGE are displayed as output. . This simple program demonstrates basic input, processing, and output operations.

Flowchart



Pseudocode

```
Begin

    numeric FIRST, SECOND, THIRD, SUM , AVERAGE

    display "Enter first number: "
    accept FIRST

    display "Enter Second number: "
    accept SECOND

    display "Enter Third number: "
    accept THIRD


    SUM  = FIRST + SECOND + THIRD

    AVERAGE  =  SUM / 3

    display FIRST
    display SECOND
    display THIRD

    display "The Sum of the 3 numbers is: ", SUM

    display "The Average of the 3 numbers is: ", AVERAGE

End
```

Link :

https://viewer.diagrams.net/?tags=%7B%7D&lightbox=1&highlight=0000ff&edit=_blank&layers=1&nav=1&title=Untitled%20Diagram.drawio&dark=auto#Uhttps%3A%2F%2Fdrive.google.com%2Fuc%3Fid%3D1gaJXk-sh5CraGSWDm3HdI_jROx-gZeA2%26export%3Ddownload

NARRATIVES
Step 1: Set Counter to 1 Step2 : Set Sum Squares to 0 Step 3: Read a card (containing a Number) Step4: If number = 0 , then go to step 9 Step 5 square the number Step6: Accumulate the sum of the squared number, i.e, add the squared number to SUM-SQAURE Step7: Increment counter, i.e. add to 1 to COUNTER Step8: Go to step 3 Step 9:Computer for the average of the sum of sqaures, i.e. divide SUM-SQuARE by COUNTER and Store in AVERAGE Step 10: print the AVERAGE Step 11: End the program

IPO

INPUT	PROCESS	OUTPUT
Counter = 1 sum_squares = 0	read or count by number of cards if number is not 0 square the number add the squared number to SUM_SQUARE increment the counter by 1 card will read again if number is 0 compuite the average by dividing the sum_sqaures and counter and store it in AVERAGE print the AVERAGE	Print the AVERAGE

FLOWCHART

PSEUDOCODE

```
begin
```

```
numeric counter = 1
numeric sum_squares = 0
numeric cards

for number by cards
      begin
            if number == 0 :
                  begin
                        squared_number = number**2
                        sum_square = squared_number + sum_square
                        counter++
                  end
            else:
                  begin
                        average = sum_sqaure/counter
                        print(average)
                        break
                  end
```

```
END
```

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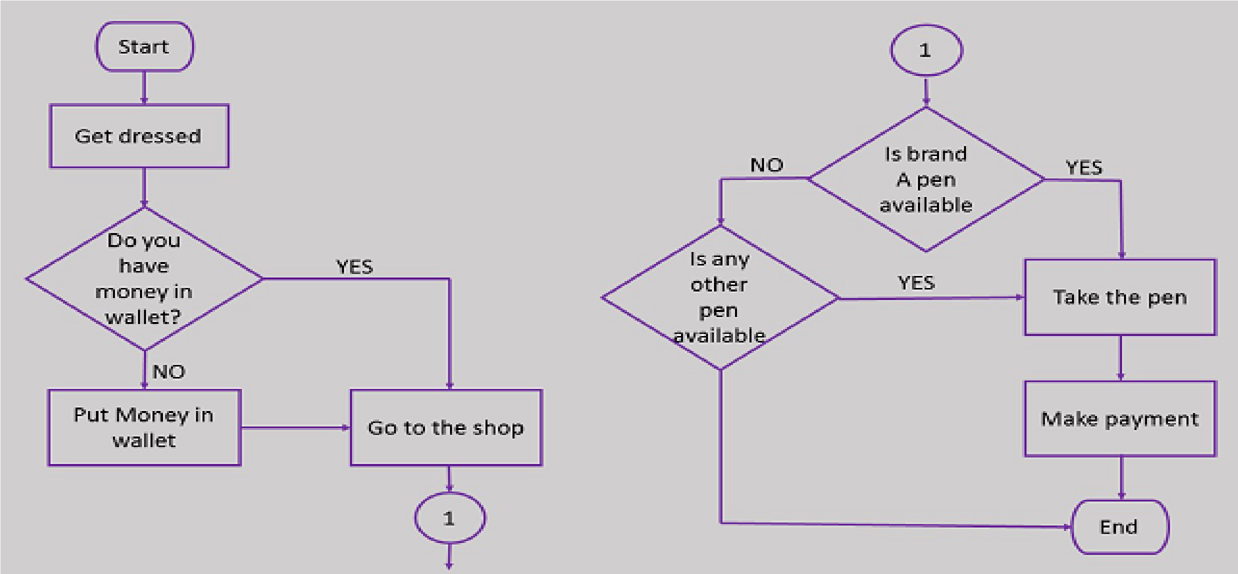
HIPO CHART

NARRATIVE DESCRIPTION

STEP 1: Get Dressed
STEP 2: Do you have money in your wallet if no go to step 3 if yes go to step 4
STEP 3: Put money in wallet
STEP 4: Go to shop
STEP 5: Is A pen brand available if no go to 6 if yes go to step 7
STEP 6: Is any other pen available if no go to 9 if yes go to step 7
STEP 7: Take the pen
STEP 8:make the payment
STEP 9: End

FLOWCHART



PSEUDOCODE

```
START
   STRING: NAME, PEN
   BOOL: DRESSED, WALLET_HAVE_MONEY, BRAND_A_AVAILABLE, OTHER_PEN_AVAILABLE

   DISPLAY "ENTER YOUR NAME"
   ACCEPT NAME

   IF DRESSED THEN
      DISPLAY "YOU GOT DRESSED"
   ELSE
      DISPLAY "CHANGE YOUR DRESS"
   ENDIF

   IF NOT WALLET_HAVE_MONEY THEN
      DISPLAY "PUT MONEY IN WALLET"
   ENDIF

   DISPLAY "GO TO SHOP"

   IF BRAND_A_AVAILABLE THEN
      PEN  "Brand A"
      DISPLAY "TAKE " + PEN
      DISPLAY "MAKE PAYMENT"
   ELSE
      IF OTHER_PEN_AVAILABLE THEN
         PEN  "Other Brand"
         DISPLAY "TAKE " + PEN
         DISPLAY "MAKE PAYMENT"
      ELSE
         DISPLAY "NO PEN AVAILABLE"
      ENDIF
   ENDIF
```

END

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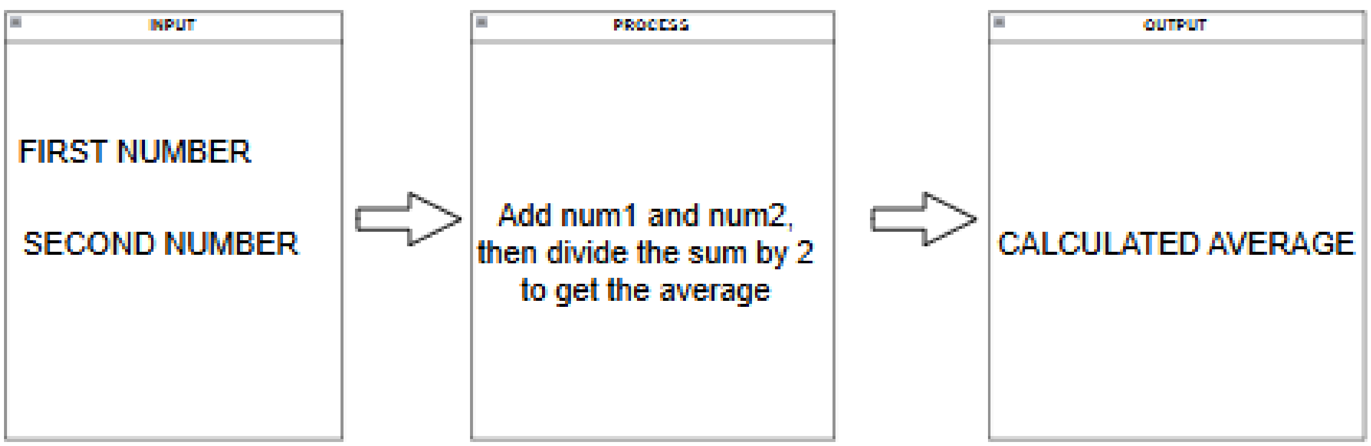
HIPO CHART



NARRATIVE DESCRIPTION

STEP 1: INPUT THE FIRST NUMBER
STEP 2: ADD NUM1 AND NUM2 THEN DIVIDE THE SUM BY 2 TO GET THE AVERAGE
STEP 3: DISPLAY THE CALCULATED AVERAGE

PSEUDOCODE

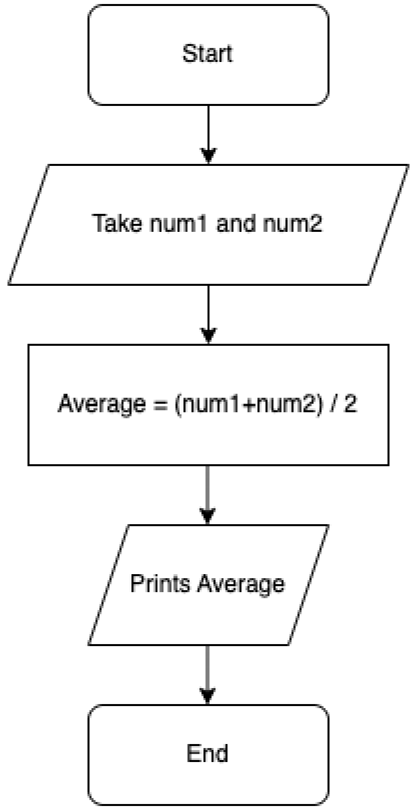
```
begin
   numeric num1, num2, nAverage
```

```
    display "ENTER THE FIRST NUMBER: "
    accept num1
    display "ENTER THE SECOND NUMBER: "
    accept num2

    nAverage = (num1 + num2) / nAverage

    display "AVERAGE: " nAverage
end
```

FLOWCHART


Start
Take num1 and num2
Average = (num1+num2) / 2
Prints Average
End

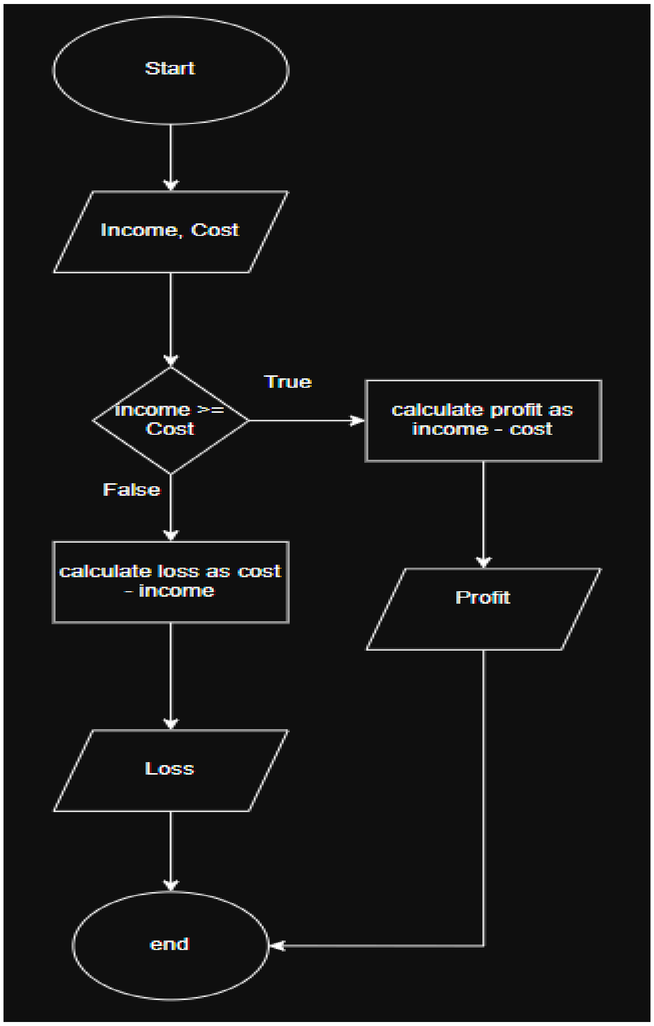
HIPO CHART

input	Process	output
input income input Cost	calculate profit as income minus cost if income greater or equal to cost if not calculate loss as cost monus income	display profit display loss

NARRATIVE DESCRIPTION

Step 1: input the Cost
step2: input the income
step3: if income greater or equal to cost
step4: calculate profit by income - cost, jump to step7
step5 if income less or equal to cost
step6:caluculate loss by cost minus income
step7:display profit
step8: display loss
step9: end

FLOWCHART


Start
Income, Cost
income >= Cost
True
calculate profit as income - cost
False
calculate loss as cost - income
Profit
Loss
end

PSEUDOCODE

```
Begin
        numeric cost, income, profit, loss
        if income >= cost:
                begin
                        profit = income - cost
                        diplay profit
                end
        else:
                begin:
                        loss = cost - income
                        display loss
                end
END
```